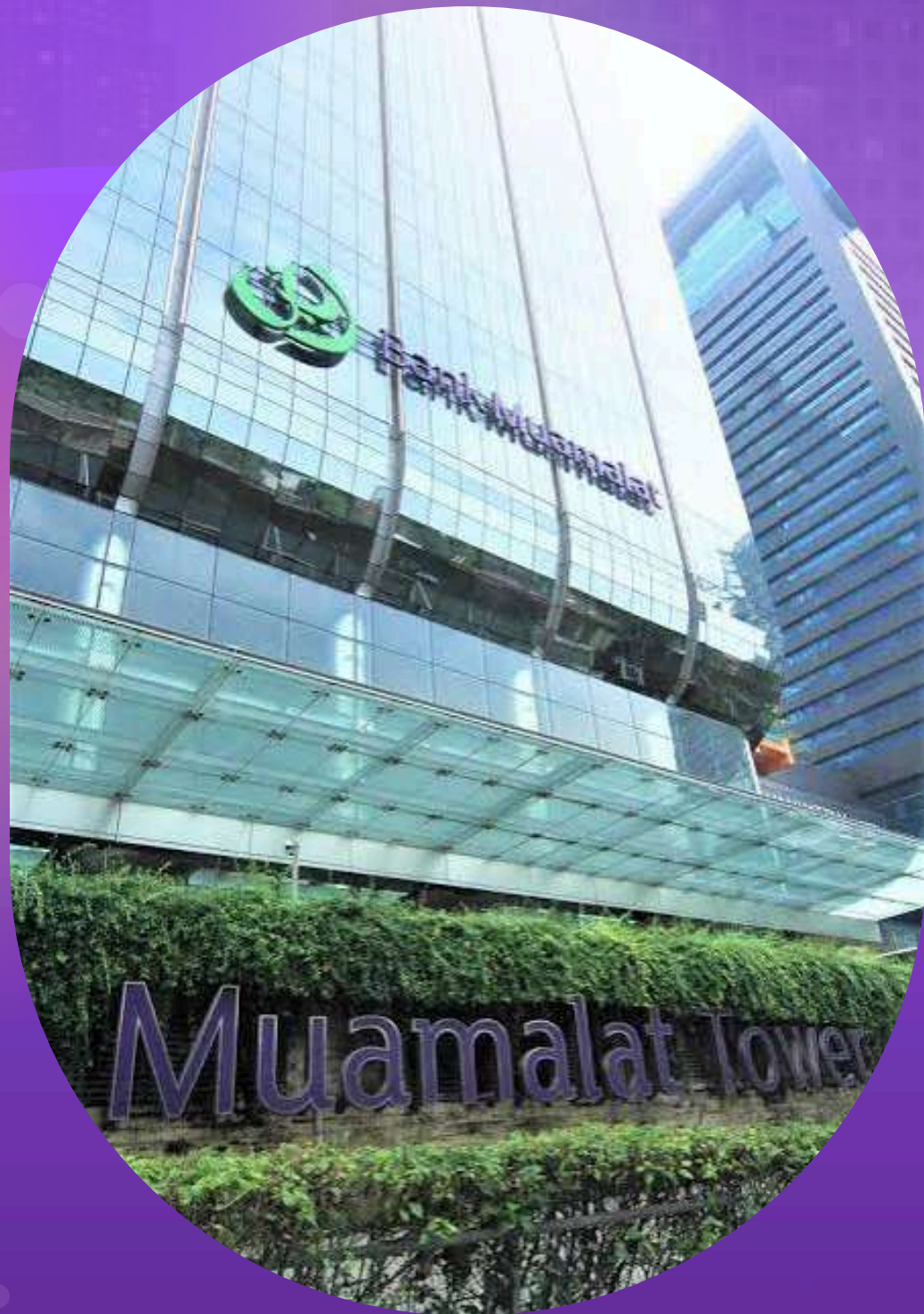




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DIGITAL USER CHURN ANALYTICS

PT SEJAHTERA BERSAMA

Bank Muamalat
Business Intelligence Analyst

Presented by : Rizky Enos Kamal





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ABOUT ME



RIZKY ENOS KAMAL

Bachelor of Statistics graduate from Universitas Brawijaya with a strong interest in Data Analytics and Business Intelligence. Experienced in data collection, processing, analysis, and visualization through various academic projects, an MBKM program, and an internship at BPS Kota Malang. Have worked on projects involving data analysis, statistical modeling, clustering, forecasting, database management, and dashboard development using Excel, Power BI, RStudio, Python, and MySQL. Possess strong analytical, attention-to-detail, communication, problem-solving, and adaptability skills, with the ability to collaborate effectively in a team.



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PROJECT OVERVIEW

OBJECTIVES

- Identify primary keys, foreign keys, and table relationships.
- Integrate four source tables into master transaction data.
- Clean and transform data using SQL in BigQuery.
- Build an interactive sales dashboard in Looker Studio.
- Generate business insights and recommendations

WORKFLOW





DATASET OVERVIEW

 Customers		
PK	<u>CustomerID</u>	<u>INTEGER</u>
	FirstName	STRING
	LastName	STRING
	CustomerEmail	STRING
	CustomerPhone	STRING
	CustomerAddress	STRING
	CustomerCity	STRING
	CustomerState	STRING
	CustomerZip	INTEGER

 Orders		
PK	<u>OrderID</u>	<u>INTEGER</u>
	Date	DATE
FK	CustomerID	INTEGER
FK	ProdNumber	STRING
	Quantity	INTEGER

 Products		
PK	<u>ProdNumber</u>	<u>STRING</u>
	ProdName	STRING
FK	Category	INTEGER
	Price	INTEGER

 ProductCategory		
PK	<u>CategoryID</u>	<u>INTEGER</u>
	CategoryName	STRING
	CategoryAbbreviation	STRING

Primary Key (PK):

- Customers → CustomerID
- Orders → OrderID
- Products → ProdNumber
- ProductCategory → CategoryID

Foreign Key (FK) :

- Orders → CustomerID, ProdNumber
- ProductCategory → Category

ENTITY RELATIONSHIP DIAGRAM (ERD)



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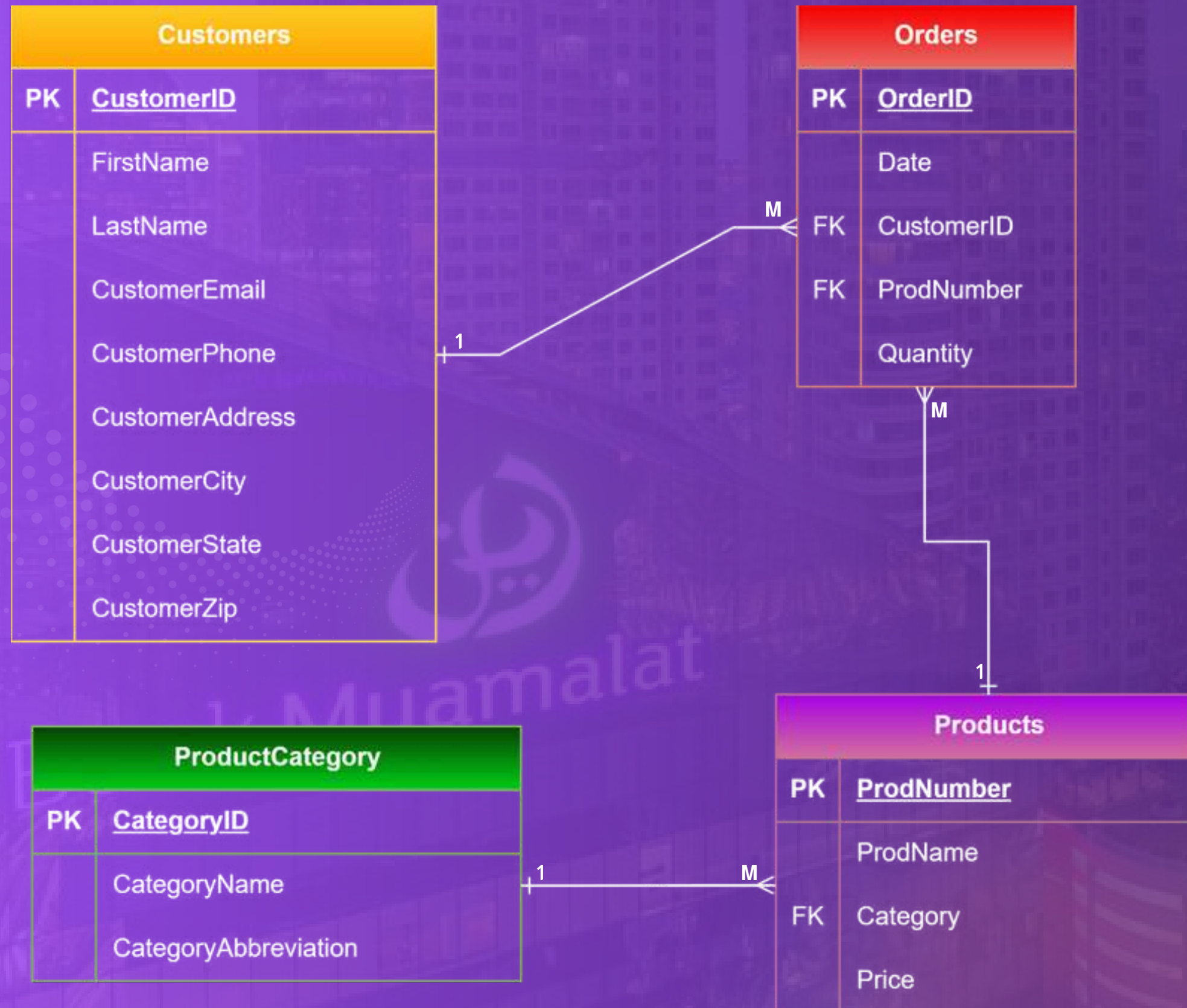


Table Relationships

- Customers → Orders: One-to-many relationship, where one customer can place multiple orders.
- Products → Orders: One-to-many relationship, where one product can appear in multiple order records.
- ProductCategory → Products: One-to-many relationship, where one category can include multiple products.

MASTER TRANSACTION DATA

```
SELECT
o.Date AS order_date,
pc.CategoryName AS category_name,
p.ProdName AS product_name,
p.Price AS product_price,
o.Quantity AS order_qty,
p.Price * o.Quantity AS total_sales,
SPLIT(c.CustomerEmail, '#')[SAFE_OFFSET(0)] AS cust_email,
c.CustomerCity AS cust_city
FROM `Dataset_BI.Orders` AS o
JOIN `Dataset_BI.Customers` AS c ON o.CustomerID = c.CustomerID
JOIN `Dataset_BI.Products` AS p ON o.ProdNumber = p.ProdNumber
JOIN `Dataset_BI.ProductCategory` AS pc ON p.Category = pc.CategoryID
ORDER BY order_date ASC;
```

Master transaction dataset containing customer, order, product, category, and sales information, then sort the transactions from the earliest to the latest order date.

Row	order_date	category_name	product_name	product_price	order_qty	total_sales	cust_email	cust_city
1	2020-01-01	eBooks	Spherical Robots	16.75	5	83.75	ibromonte9@de.vu	Birmingham
2	2020-01-01	Training Videos	Drone Video Techniques	37.99	6	227.94	gstiggersdd@eventbrite.com	Saint Petersburg
3	2020-01-01	Drone Kits	BYOD-220	69.0	1	69.0	edee@nba.com	Honolulu
4	2020-01-01	eBooks	SCARA Robots	19.5	5	97.5	bespercx@com.com	Des Moines
5	2020-01-01	eBooks	Polar Robots	23.99	2	47.98	fraserlqz@comsenz.com	Jackson
6	2020-01-01	Robots	BWW-75 Robot	883.0	3	2649.0	tmckamot@tinyurl.com	Katy
7	2020-01-02	eBooks	Fixed Wing Drones	15.5	3	46.5	chaiberdv@booking.com	Birmingham
8	2020-01-02	Training Videos	Understanding Automation	44.95	1	44.95	ksteenshp@ameblo.jp	San Diego
9	2020-01-02	Drones	DTE-QFN20 Drone	250.0	2	500.0	jcolthurstga@cbsnews.com	Sacramento
10	2020-01-02	Robot Kits	BYOR-2640S	189.0	2	378.0	qminteres4v@state.tx.us	Washington

SALES PERFORMANCE DASHBOARD



Key Performance Indicators (KPIs)

Sales Trend & Yearly Comparison

Product & City Performance

Interactive Filters

Product Category Ranking

Product Category Performance



INTERACTIVE FILTERS

Date Range Filter

Jan 1, 2020 - Dec 31, 2021 ▾

Fixed ▾

Start Date End Date

JAN 2020 ▾ < > DEC 2021 ▾ < >

S M T W T F S S M T W T F S

JAN 1 2 3 4 DEC 1 2 3 4

5 6 7 8 9 10 11 5 6 7 8 9 10 11

12 13 14 15 16 17 18 12 13 14 15 16 17 18

19 20 21 22 23 24 25 19 20 21 22 23 24 25

26 27 28 29 30 31 26 27 28 29 30 31

Cancel Apply

Allows users to analyze sales performance within a selected period and identify changes or fluctuations over time.

Product Category Filter

Product Category ▾

✓ Product Category	Record Count
🔍 Type to search	
✓ eBooks	891
✓ Training Videos	615
✓ Robots	291
✓ Robot Kits	302
✓ Drones	352
✓ Drone Kits	433
✓ Blueprints	455

Enables users to evaluate the contribution and performance of specific product categories.

KEY PERFORMANCE INDICATORS (KPIs)



Total Sales
\$1.75M

The company generated approximately \$1.75M in total sales during the analyzed period, reflecting the overall revenue contribution from all transactions.



Total Quantity
11.65K

Approximately 11.65K units were sold, indicating the overall product demand and transaction volume.



Total Transaction
3,339

A total of 3,339 transactions were recorded, showing the level of purchasing activity during the period.



Average Sales
\$525.53

Each transaction generated an average sales value of approximately \$525.53, indicating the typical revenue contribution per transaction.



Total Customer
1.67K

Approximately 1.67K unique customers contributed to sales, representing the active customer base during the analyzed period.



SALESTREND & YEARLY COMPARISON

Monthly Sales Comparison



Yearly Comparison

2020
\$913.21K

2021
\$841.54K
-7.8%

Positive Spike

June 2021 recorded the highest sales peak during the analysis period, driven by higher transaction activity and a stronger contribution from high-value categories, particularly Robots and Drones

Sales Volatility

Sales in 2021 fluctuated more strongly than in 2020, indicating inconsistent monthly performance and a need for closer demand monitoring.

Long-Term Challenge

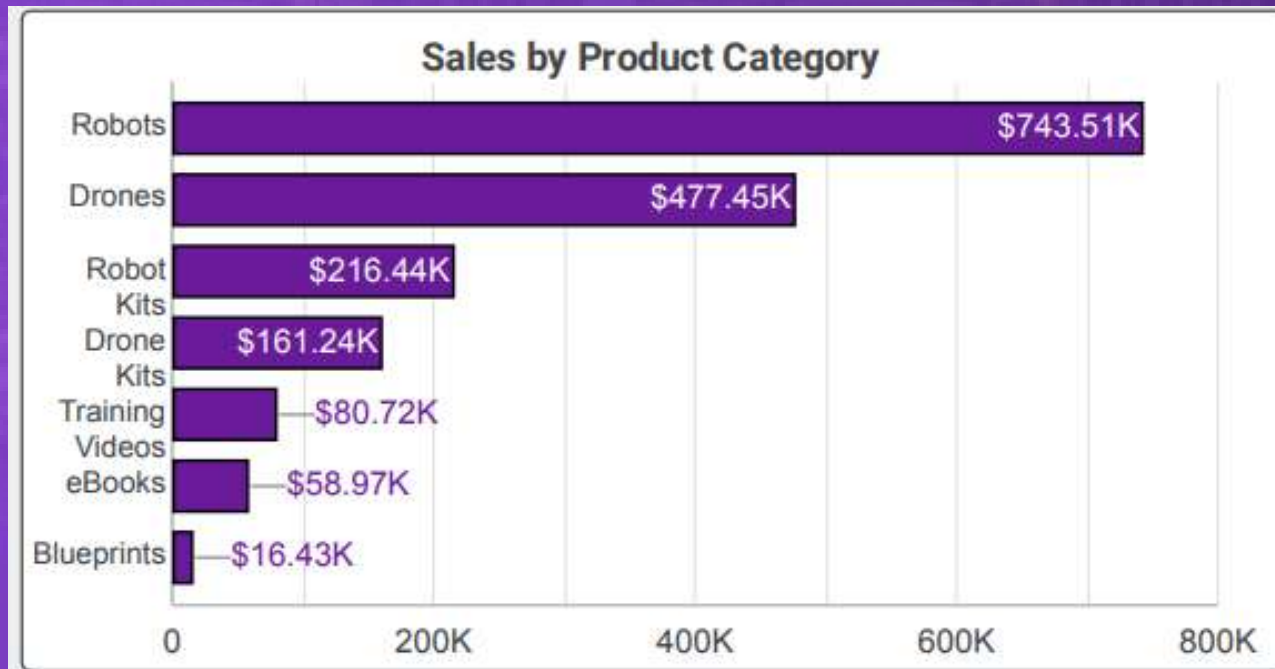
Despite the strong sales peak in June, annual sales still declined by 7.8% YoY, indicating that short-term gains were not enough to offset weaker performance in other periods.



PRODUCT CATEGORY PERFORMANCE

TOP 5 PRODUCT CATEGORY BY SALES

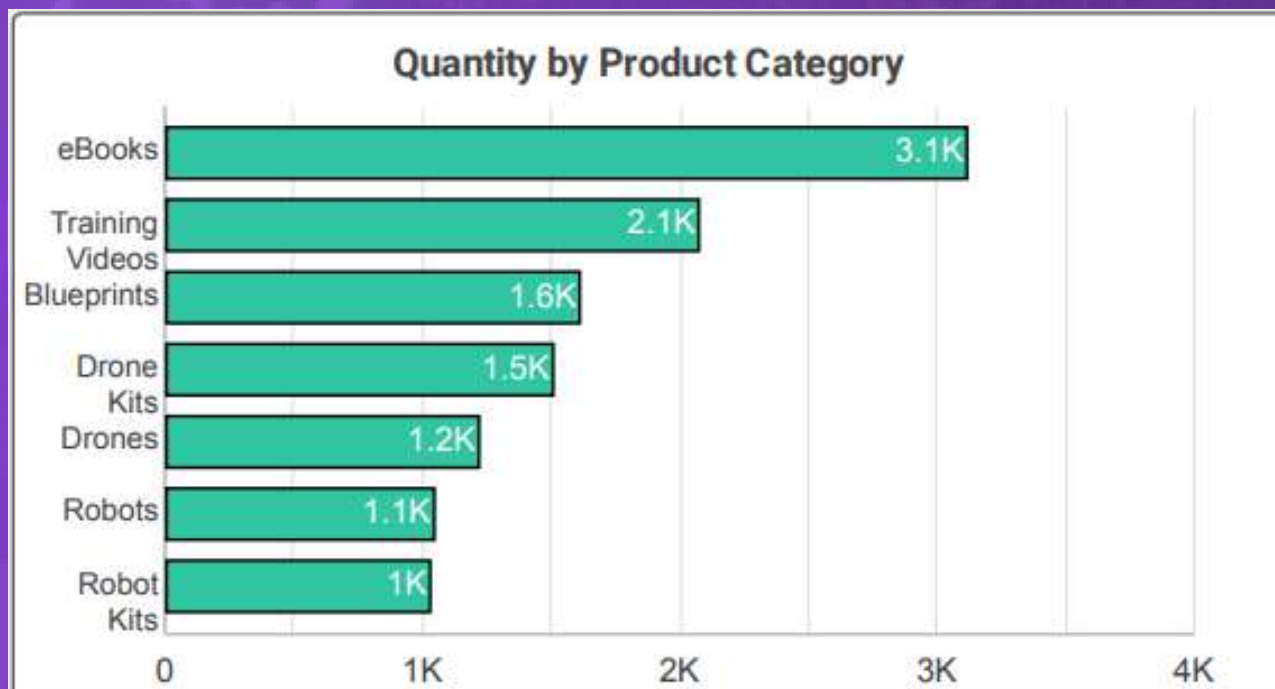
1	Robots	\$743.51K
2	Drones	\$477.45K
3	Robot Kits	\$216.44K
4	Drone Kits	\$161.24K
5	Training Videos	\$80.72K



- Robots and Drones dominate revenue contribution, accounting for nearly 70% of total sales, indicating a strong dependence on high-value product categories.
- Maintaining the sales of these categories is strategically important, as any decline could significantly impact overall revenue.

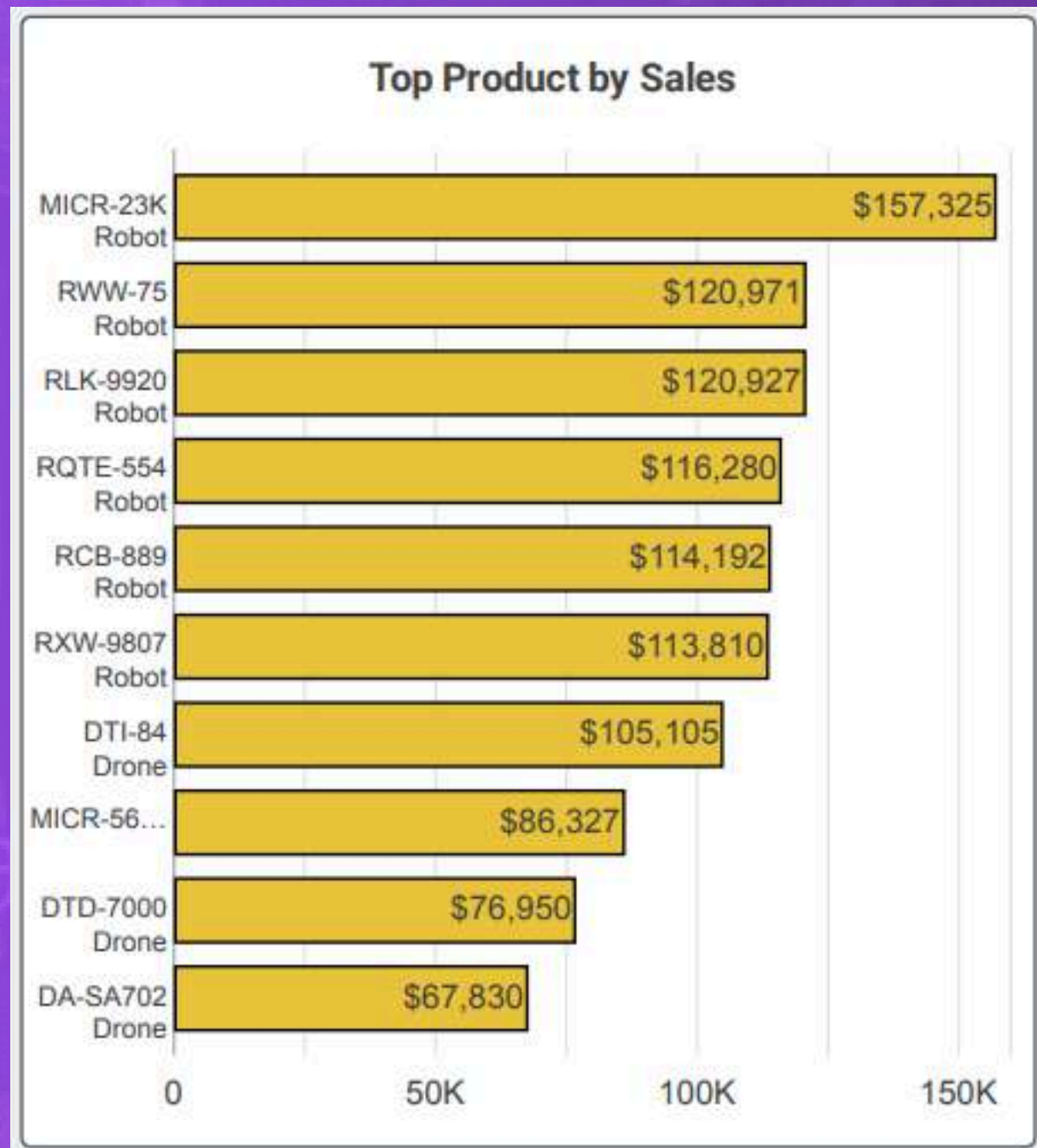
TOP 5 PRODUCT CATEGORY BY QUANTITY

1	eBooks	3.1K
2	Training Videos	2.1K
3	Blueprints	1.6K
4	Drone Kits	1.5K
5	Drones	1.2K



- eBooks and Training Videos lead in quantity sold, indicating strong demand for lower-value digital products despite their smaller revenue contribution.
- Their high sales volume creates opportunities for bundling and cross-selling with higher-value products to increase average transaction value.

PRODUCT PERFORMANCE

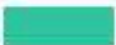


Robot products dominate the top sales ranking, with MICR-23K Robot generating the highest sales at approximately \$157.33K. This concentration highlights the strategic importance of maintaining the performance and availability of top Robot products.



CITY PERFORMANCE



	City	Sales Revenue	Quantity ▾
1.	Washington	\$55.38K 	308 
2.	Houston	\$33.76K 	249 
3.	San Diego	\$29.23K 	203 
4.	Atlanta	\$18.98K 	153 
5.	Sacramento	\$33.38K 	153 
6.	Chicago	\$15.53K 	152 
7.	Springfield	\$25.06K 	149 
8.	Denver	\$20.25K 	141 
9.	San Antonio	\$18.88K 	140 
10.	Philadelphia	\$23.85K 	139 

1 - 10 / 361 < >

Washington leads both sales and quantity, while cities such as Sacramento generate relatively high revenue despite lower sales volume. This difference between sales value and quantity indicates opportunities to apply location-specific sales strategies based on the characteristics of each market.



BUSINESS RECOMMENDATIONS



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Protect Revenue Drivers

Prioritize product availability and sales support for Robots and Drones, which contribute nearly 70% of total sales.



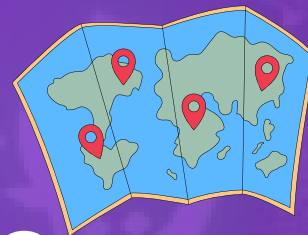
Increase Transaction Value

Bundle high-volume products such as eBooks and Training Videos with higher-value products to increase average transaction value.



Strengthen Customer Retention

Use customer transaction history and email data to support personalized retention, repeat-purchase, and re-engagement campaigns.



Optimize Geographic Strategy

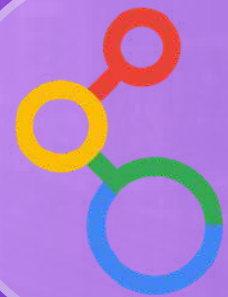
Apply city-specific sales strategies based on differences in revenue, sales volume, and customer purchasing patterns.



Monitor Sales Performance

Analyze the drivers behind the June 2021 sales peak and continuously monitor weaker periods to address the 7.8% YoY sales decline.

PROJECT LINK



Looker Studio
Dashboard



Video
Presentasi



Link Github

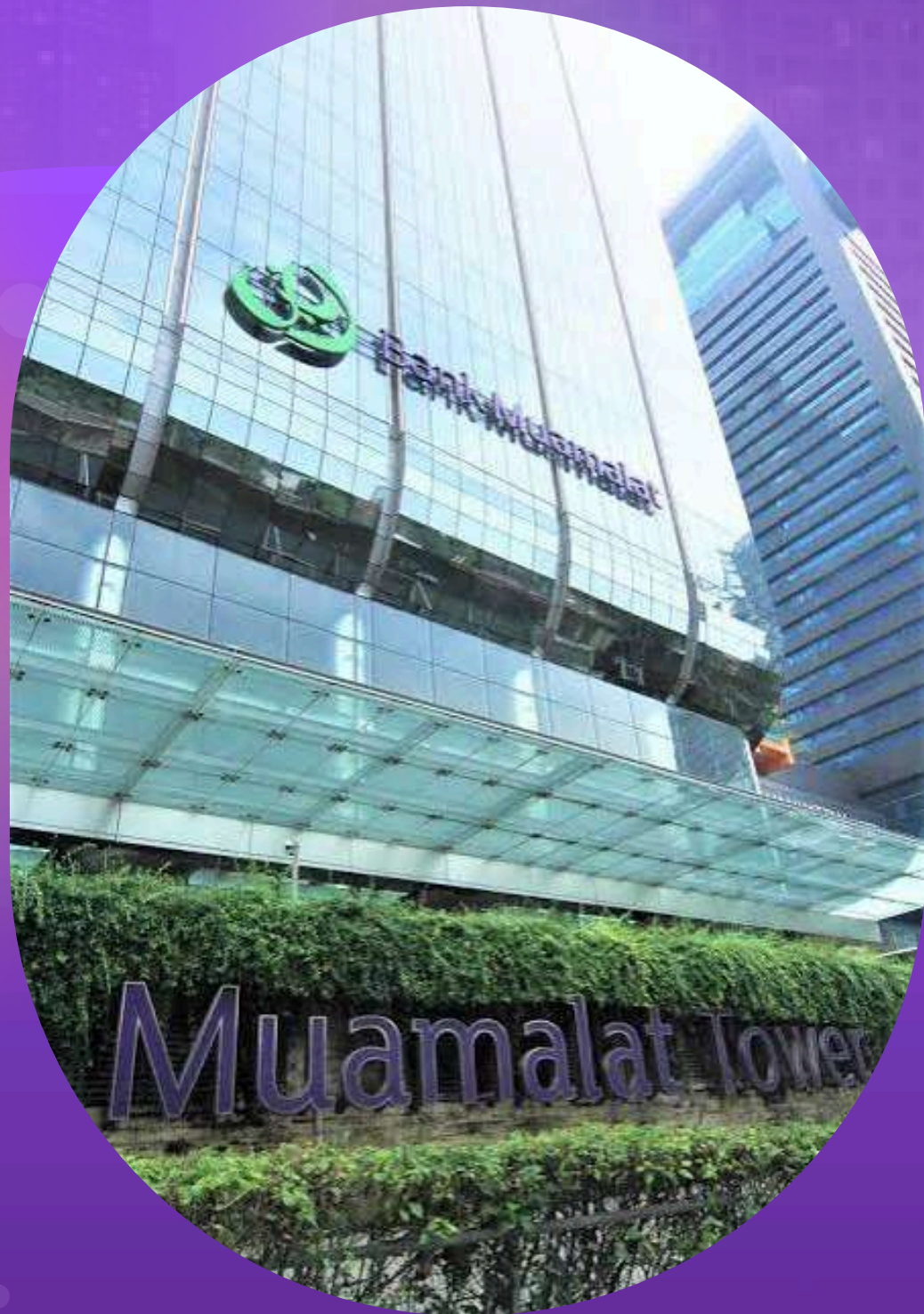




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